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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)**Second Semester M.Tech (Electric Vehicle Technology) (CBCS) Degree Examinations**

May - June 2024

22EVT201 – POWER ELECTRONIC CONVERTERS

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Draw the characteristics of an ideal power transistor and explain the same.	6	L*1	1	1
b)	With analysis, and explanation, obtain the typical switching wave forms for the Power delivery.	8	L2	1	2
c)	Draw the structure and the equivalent circuit of the power Schottky rectifier and justify that why it is unipolar device?	6	L1	1	1
2. a)	With neat circuit diagrams and typical waveforms, explain the operation of a typical buck converter for continuous conduction mode with analysis.	10	L2	1	2
b)	With the aid of relevant boost converter circuit configuration and typical waveforms, explain the on-time event and off-time event boost converter for continuous conduction mode of operation with analysis.	10	L1	1	1
Unit – II					
3. a)	Obtain the analysis for A 12 V, 4 A Voltage-Mode Buck from a 28 V Source. The detail specifications are as below: The specification states that the output must be 12 V when loaded by a 4 A current and powered by a source varying from 20 V to 30 V. The maximum peak-to-peak output voltage ripple must always stay below 125 mV, and the switching frequency is 100 kHz. Please note an ac input ripple current specification of 15 mA peak to peak.	10	L2	3	4
b)	Obtain the analysis for a Voltage-Mode 12 V, 2 A Buck-Boost Converter Powered from a Car Battery. The detail specifications are as below: $V_{in, min} = 10\text{ V}$; $V_{in, max} = 15\text{ V}$; $V_{out} = -12\text{ V}$; $V_{ripple} = \Delta V = 250\text{ mV}$; $V_{out\ drop} = 250\text{ mV}$ maximum from $I_{out} = 0.2\text{ to }2\text{ A}$ in $10\text{ }\mu\text{s}$; $I_{out, max} = 2\text{ A}$; $F_{sw} = 100\text{ kHz}$.	10	L3	3	4
4. a)	Obtain the design analysis for a 48 V, 2 A, Voltage - Mode Boost from a car battery. A boost converter is designed to deliver around 100 W from a 12 V car battery. The detail specifications are as below: $V_{in, min} = 10\text{ V}$; $V_{in, max} = 15\text{ V}$; $V_{out} = 48\text{ V}$; $V_{ripple} = \Delta V = 250\text{ mV}$; $V_{out\ drop} = 200\text{ mV}$ maximum from $I_{out} = 1\text{ to }10\text{ A}$ in $10\text{ }\mu\text{s}$; $I_{out, max} = 2\text{ A}$; $F_{sw} = 300\text{ kHz}$.	12	L3	3	4
b)	Draw the neat ripple current wave form in the CCM buck inductor with details.	8	L2	2	2
Unit – III					
5. a)	Obtain the technical note on Graetz bridge with a help of neat circuit diagram and obtain the expression for the followings: (a) Input voltage, (b) Depletion current, (c) Peak recharge capacitor voltage.	10	L4	3	4
b)	Obtain the expression for Power Factor when the average input power drawn by a load actually corresponds to the instantaneous power $p_{in}(t)$ averaged over one cycle.	10	L4	4	4

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| 6. a) | Explain in detail the different techniques adopted for Active Power Factor Correction. | 10 | L4 | 3 | 5 |
| b) | Obtain the expression for the average diode current and power dissipated in the diode. | 10 | L4 | 4 | 4 |

Unit – IV

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| 7. a) | Obtain the analysis for Flyback converter without parasitic elements in CCM with help of neat circuit diagrams, wave forms and explanation. | 10 | L4 | 4 | 5 |
| b) | Obtain the analysis for Flyback converter with parasitic elements in CCM with help of neat circuit diagrams, wave forms and explanation. | 10 | L4 | 4 | 5 |
| 8. a) | With the help of SPICE circuit diagram, obtain the technical note and analysis of Clamping the Drain Excursion of the network. | 10 | L5 | 4 | 4 |
| b) | Explain Two-Switch Flyback Converter with the help of neat circuit diagram. | 10 | L3 | 4 | 4 |

Unit – V

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| 9. a) | With neat operating circuit modes and analysis explain the operation of a bidirectional dc-dc converter used in EV. Also obtain average state space model of the same. | 10 | L5 | 5 | 5 |
| b) | Draw the neat sketches of any six types of different bidirectional ac-dc converter topologies as applicable in PHEV for battery charging. | 10 | L5 | 5 | 4 |
| 10. a) | Write a technical note on the followings: - | | | | |
| | 1. Two stage ac-dc converter with inductive power transfer | | | | |
| | 2. Full-bridge bidirectional single stage ac-dc converter | 10 | L5 | 3 | 5 |
| b) | Obtain the any five comparison of different ac-dc converter used in PHEV. | 10 | L5 | 5 | 4 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
